EE619 VLSI System Design 2025-26 Sem-II

Assignment 1 **Dated: 15/01/2026** **Due Date : 28/01/2026**

Instructions:

1. Please use Virtuoso with gpdk180 transistor models. Refer to RV_CMOS_INV_Schematic_design.mp4 to learn how to do this. The path to be added to the library path editor is /home/vlsi_IITK/GPDK/gpdk180_v3.3/libs.oa22/gpdk180.
2. Please refer to the pdf 'EE619_Cadence_setup.pdf' for instructions on the tool setup.

Q1. Simulate the output characteristics for various values of V_{GS} for an NMOS transistor.

(a) Plot output characteristics for an NMOS transistor (W = 1 µm, L = 500 nm). Vary V_{DS} from 0 V to 1.8 V in steps of 0.1 V. Plot this for V_{GS} = {0 V, 0.6 V, 1.2 V, 1.8 V}. Obtain these characteristics for V_{SB} = 0 V.

(b) Plot output conductance for an NMOS transistor (W = 1 µm, L = 500 nm). Vary V_{DS} from 0 V to 1.8 V in steps of 0.1 V. Plot this for V_{GS} = {0 V, 0.6 V, 1.2 V, 1.8 V}. Obtain these characteristics for V_{SB} = 0 V.

Q2. Simulate the transfer characteristics for various values of V_{DS} for an NMOS transistor.

(a) Plot transfer characteristics for an NMOS transistor (W = 1 µm, L = 500 nm). Vary V_{GS} from 0 V to 1.8 V in steps of 0.1 V. Plot this for V_{DS} = {0 V, 0.6 V, 1.2 V, 1.8 V}. Obtain these characteristics for V_{SB} = 0 V.

(b) Plot transconductance for an NMOS transistor (W = 1 µm, L = 500 nm). Vary V_{GS} from 0 V to 1.8 V in steps of 0.1 V. Plot this for V_{DS} = {0 V, 0.6 V, 1.2 V, 1.8 V}. Obtain these characteristics for V_{SB} = 0 V.

(c) Plot transfer characteristics for an NMOS transistor (W = 1 µm, L = 500 nm). Vary V_{GS} from 0 V to 1.8 V in steps of 0.1 V. Plot this for V_{DS} = {0 V, 0.6 V, 1.2 V, 1.8 V}. Obtain these characteristics for V_{SB} = 0.5 V.

Q3. Plot I_{SD} vs V_{SD} for a PMOS transistor (W = 1 µm, L = 500 nm). Vary V_{SD} from 0 V to 1.8 V in steps of 0.1 V. Plot this for V_{SG} = {0 V, 0.6 V, 1.2 V, 1.8 V}. Obtain these characteristics for V_{SB} = 0 V.

Q4. Plot I_{SD} vs V_{SG} for a PMOS transistor (W = 1 µm, L = 500 nm). Vary V_{SG} from 0 V to 1.8 V in steps of 0.1 V. Plot this for V_{SD} = {0 V, 0.6 V, 1.2 V, 1.8 V}. Obtain these characteristics for V_{SB} = 0 V.

Q5. Open the model file you have used for the transistors while simulation. (/home/vlsi_IITK/GPDK/gpdk180_v3.3/models/spectre/nmos1.scs for the NMOS transistor model.) Observe the contents of the file.

Submission Guidelines:

File Name: Rollno._Assign1_EE619.pdf

- Page 1:
  - Please add your name and roll number in the top right corner.
  - Fig. 1: Output characteristics for an NMOS transistor as described in Q1 (a).
  - Fig. 2: Output conductance for an NMOS transistor as described in Q1 (b).
  - Observation 1: Comments on the Fig. 2., and your observations.
- Page 2:
  - Fig. 3: Transfer characteristics for an NMOS transistor as described in Q2 (a).
  - Fig. 4: Transconductance for an NMOS transistor as described in Q2 (b).
  - Observation 2: Comments Fig. 4., and your observations.
- Page 3:
  - Fig. 5: I_{DS} vs V_{GS} for an NMOS transistor as described in Q2 (c).
  - Observation 3: Compare Fig. 3. and Fig. 5. and note down your observations.
- Page 4:
  - Fig. 6: I_{SD} vs V_{SD} for an PMOS transistor as described in Q3.
  - Fig. 7: I_{SD} vs V_{SG} for an PMOS transistor as described in Q4.

Note:

- Please make sure that your plots are readable. X-Y axis labels and values should be readable, the legend should be clear, plots should have reasonable aspect ratio, size etc. If you find it difficult to do this in Virtuoso directly, you can export the plots as *.csv file. This file can then be read into Excel, Matlab (you have access to Matlab online), your own custom Python script, etc. to generate a clean plot. Establish a clean way to get plots at this stage. This will be very handy in your future submissions.